

BCI-FES Integration for Motor Rehabilitation: A Scoping Review

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Abstract—Operational integration of brain-computer interfaces (BCIs) and functional electrical stimulation (FES), in which real-time neural signals trigger muscle activation, offers a neurobiologically grounded route to motor recovery, yet its technical scope has not been systematically mapped. We conducted a scoping review following PRISMA-ScR, searching PubMed, Scopus, and IEEE Xplore through March 2026. Eighty-two studies (2010–2025) were included. Stroke accounted for 72% of the clinical population and spinal cord injury for 28%; non-invasive EEG was dominant (94%), with motor imagery (48%) and movement execution or attempt (38%); surface FES was used in 93%. Twenty-nine percent measured only system performance, without a clinical outcome. Of the 26 studies (32%) with a separate control group, 17 compared groups on a clinical outcome, with BCI-FES superior in 10 (9 of 16 independent cohorts), a vote count across mostly underpowered studies (median $n=7$) that should be read as a descriptive signal, not a measure of efficacy. Randomized trials numbered 15 (18%). FES frequency and pulse width were unreported in 41% and 39% of studies. Technical feasibility is suggested mainly in chronic upper-limb stroke, while efficacy evidence rests on a few small controlled studies and FES parameter reporting remains variable.

Index Terms—brain-computer interface, functional electrical stimulation, motor rehabilitation, electroencephalography, scoping review

I. INTRODUCTION

Stroke and spinal cord injury (SCI) are among the leading causes of permanent motor disability worldwide, imposing a substantial burden on individuals, caregivers, and health systems [1], [2]. Conventional rehabilitation approaches (physiotherapy, occupational therapy, task-oriented training) have shown limited efficacy in patients with severe deficits, motivating the development of neurotechnology-assisted interventions [3].

Brain-computer interfaces (BCIs) capture neural signals in real time, most commonly through electroencephalography

(EEG), and translate them into commands for external devices, bypassing injured motor pathways [4]. Functional electrical stimulation (FES) delivers controlled electrical current to peripheral nerves or muscles, eliciting functional contractions in otherwise paralyzed limbs [5]. Both modalities have been investigated independently as rehabilitation tools; each, however, addresses only part of the neural-motor circuit.

When decoded cortical signals trigger the delivery of FES in real time, the result is a closed neuromuscular loop grounded in the principles of Hebbian plasticity. The simultaneous activation of the motor cortex and the peripheral muscle during intentional motor effort is hypothesized to strengthen corticospinal connections and to support long-term motor recovery [6], [7]. This rationale has sustained a growing body of experimental work over the past 15 years.

Existing systematic reviews and meta-analyses have examined the efficacy of BCI-FES in upper-limb rehabilitation after stroke [8]–[10]. This scoping review complements that evidence by characterizing the broader landscape: the populations studied, the technical configurations used, and the outcomes measured across all study designs, providing context for interpreting and extending the evidence from controlled trials. Previous reviews, including systematic and narrative syntheses focused on efficacy in specific populations [8]–[10], concentrated on the subset of controlled trials; to our knowledge, no prior scoping review has characterized the broader technical landscape of BCI-FES integration, spanning all study designs and populations, under the PRISMA-ScR framework.

This review aims to characterize study designs, populations, technical BCI and FES parameters, and reported outcomes in the BCI-FES integration literature, identifying patterns and gaps that may inform future research and reporting guidelines.

II. METHODS

A. Study Design

This scoping review was conducted following the PRISMA extension for scoping reviews (PRISMA-ScR) [11], with the protocol, eligibility criteria, and search strategy defined a priori.

B. Search Strategy

Three electronic databases were searched through March 20, 2026. PubMed was selected for its clinical and biomedical coverage; Scopus for its interdisciplinary breadth; and IEEE Xplore for its engineering and neurotechnology publications, where a substantial share of BCI research appears and is not fully indexed in the other databases. The search combined terms related to brain-computer interfaces (“brain-computer interface”, “BCI”, “brain-machine interface”, “BMI”), functional electrical stimulation (“functional electrical stimulation”, “FES”, “neuromuscular electrical stimulation”, “NMES”), and motor rehabilitation (“motor rehabilitation”, “motor recovery”, “neuromotor rehabilitation”). No language restriction was applied, and the strategy was adapted to the syntax of each database. The full search string is available from the authors on request.

C. Inclusion and Exclusion Criteria

Studies were included if they: (I1) involved adults (≥ 18 years) with motor deficits secondary to a neurological condition; (I2) implemented real-time BCI-FES integration, defined as central neural signals captured by a BCI directly and continuously modulating FES delivery; (I3) reported at least one quantitative outcome from an implemented system; and (I4) were published between January 2010 and March 20, 2026, as full papers.

Studies were excluded if they: (E1) involved exclusively healthy participants; (E2) used BCI and FES in temporally separate phases, without real-time integration; (E3) controlled FES through electromyography, movement sensors, or manual control without an operational neural BCI; (E4) used a powered robotic exoskeleton as the primary effector without FES; (E5) were exclusively computational studies, simulations, or animal experiments; or (E6) were reviews, meta-analyses, editorials, protocols without collected data, or conference abstracts without a full paper.

D. Screening and Data Extraction

Titles and abstracts of 1,161 unique records were screened by a single reviewer; 168 full texts were assessed for eligibility. Single-reviewer screening is an acknowledged limitation, mitigated by requiring a supporting verbatim quotation from the source article for the six classification fields most subject to interpretation (design, control group, BCI paradigm, BCI-FES integration, outcome, and between-group superiority). Data were extracted using a standardized form with 52 fields across eight domains: study identification, participants, study design, BCI characteristics, FES characteristics, BCI-FES integration,

intervention protocol, and outcomes. Categorical fields followed a closed vocabulary; any field without explicit information in the source article was coded “NR” (not reported), and inference was not permitted. The unit of analysis was the individual publication. Some groups contributed multiple articles, and a subset reports overlapping or identical participant cohorts: the Remsik et al. series (2018, 2019, 2021, 2022) [55], [60], [73], [80], derived from a common cohort; two publications from the same Krueger et al. cohort (2022, 2024) [15], [78]; and two publications on the same patient by Bastos-Filho et al. [92], [93]. Participant totals should therefore be read at the report level, without assuming strictly independent samples.

E. Classification Criteria

To reduce subjectivity, each categorical field followed an explicit rule. Design was classified hierarchically: randomized trial (≥ 2 groups with random allocation), non-randomized controlled trial (≥ 2 groups without randomization), single case ($n=1$), and, for single-arm samples with $n \geq 2$, feasibility/pilot study (when the stated primary outcome was feasibility, accuracy, or safety) or case series (when it described clinical outcomes). The control-group field was coded as present only when a separate, prospectively allocated participant group existed; within-subject comparisons (e.g., A-B-A-B or condition-crossover designs) and historical controls were treated as the absence of a control group. The outcome was labeled according to what the study set out to measure: studies without any pre/post clinical comparison were coded “no clinical outcome”, regardless of technical performance; the remainder were classified as positive, mixed, or negative by the pre-specified clinical outcome, based on the Results section rather than the Discussion. Superiority was assigned only when a study reported a between-group statistical test on a clinical outcome.

III. RESULTS

A. Study Selection

The database searches identified 1,301 records (PubMed: 118; Scopus: 981; IEEE Xplore: 202). After deduplication, 1,161 unique records were screened at the title and abstract level; 194 were sought for retrieval, of which 26 were not retrieved, yielding 168 articles assessed in full text. Of these, 82 met all inclusion criteria (Fig. 1). The most common reasons for full-text exclusion were reviews or records without primary data ($n=24$), the inclusion of healthy participants only ($n=23$), and FES driven by electromyographic, sensor, or manual control without an operational BCI ($n=12$); the full breakdown is shown in Fig. 1. Table III summarizes the 82 studies individually.

B. Publication Trends

The included studies were published between 2010 and 2025. Thirty-five percent of the corpus ($n=29$) appeared in the last five years (2021–2025), reflecting sustained growth in publication volume across the review period.

C. Study and Population Characteristics

Study designs were heterogeneous (Table I). Designs without a separate control group predominated: single-case studies ($n=19$, 23%), feasibility or pilot studies without control ($n=15$, 18%), and case series ($n=10$, 12%). Among designs with a comparator, non-randomized controlled trials accounted for 20 studies (24%), some using within-subject or historical comparison, and randomized trials for 15 (18%, including 3 randomized pilots); three studies had other designs (a comparison including healthy adults, a technical parameter report, and a case study). In total, 26 studies (32%) included a separate (between-subjects) control group, the denominator adopted for the superiority analyses below.

Stroke was the predominant clinical condition ($n=59$, 72%); SCI accounted for 28% ($n=23$). Among the 72 studies that reported the clinical phase, participants in the chronic phase were the most common ($n=47$, 65%), followed by the subacute phase ($n=17$, 24%), mixed-phase populations ($n=7$, 10%), and the acute phase ($n=1$, 1%).

Upper-limb rehabilitation was the primary focus in 78% of studies ($n=64$); lower-limb targets were addressed in 16% ($n=13$) and both segments in 5% ($n=4$). Sample sizes, defined as the number of participants recruited, ranged from 1 to 64 (median: 7; mean: 10.3). Twenty studies (24%) involved a single participant. Table I summarizes the corpus characteristics.

TABLE I
CHARACTERISTICS OF THE 82 INCLUDED STUDIES

Characteristic	Category	n (%)
Design	Single case	19 (23%)
	Feasibility/pilot, no control	15 (18%)
	Case series	10 (12%)
	Non-randomized controlled	20 (24%)
	Randomized trial	15 (18%)
	Other	3 (4%)
Clinical condition	Stroke	59 (72%)
	Spinal cord injury	23 (28%)
Clinical phase^a	Chronic	47 (65%)
	Subacute	17 (24%)
	Mixed	7 (10%)
	Acute	1 (1%)
Target limb	Upper	64 (78%)
	Lower	13 (16%)
	Both	4 (5%)
	Not reported	1 (1%)
BCI type	Non-invasive EEG	77 (94%)
	Invasive (intracortical)	5 (6%)
BCI paradigm	Motor imagery	39 (48%)
	Execution/attempt	31 (38%)
	ERD/MRCP, attention, P300	11 (13%)
	Other	1 (1%)
FES type	Transcutaneous	76 (93%)
	Peripheral nerve	1 (1%)
	Percutaneous	1 (1%)
	Not reported	4 (5%)
Outcome	Positive	46 (56%)
	Mixed	12 (15%)
	No clinical outcome	24 (29%)

^aPhase percentages computed over the 72 studies reporting phase.

D. BCI Characteristics

Non-invasive EEG was the dominant BCI modality, used in 77 of the 82 studies (94%). Fully implanted intracortical arrays appeared in 5 studies (6%), aimed mainly at restoring

grasp and reach in SCI [14]. Motor imagery (MI), in which participants imagined limb movement without physical execution, was the most common paradigm ($n=39$, 48%), followed by movement execution or attempt (ME), in which participants actively tried to perform the movement ($n=31$, 38%). MRCP/ERD-based paradigms without explicit MI/ME classification were used in 6 studies (7%), generally aimed at inducing associative neuroplasticity [18]; attention/concentration indices and P300 appeared in the remainder.

Recording channels were reported in 90% of studies ($n=74$), with a median of 16; the highest values correspond to the invasive intracortical studies. Electrode placement predominantly targeted the sensorimotor cortex: C3, C4, and Cz (the canonical hand and arm motor cortex positions in the 10-20 system) appeared in the large majority of configurations, often supplemented by frontocentral and centroparietal electrodes; frontal-pole electrodes (Fp1/Fp2) appeared in only 8 studies (10%). EEG frequency bands were reported in 90% of studies ($n=74$); the mu rhythm (8–13 Hz) and the beta band (13–30 Hz) were the primary features, with most studies applying broadband 8–30 Hz filtering that captures both. Classification algorithms were reported in 94% of studies ($n=77$), with LDA, threshold-based ERD detection, and SVM among the most frequent.

E. FES Characteristics

Transcutaneous FES was used in 76 studies (93%), intramuscular percutaneous FES in 1 (1%), peripheral nerve stimulation in 1 (1%), and 4 studies (5%) did not report the stimulation type. Among the 48 studies that reported the stimulation frequency, the median was 35 Hz; the parameter was not reported in 41% of studies ($n=34$). Pulse width had a median of 250 μ s among the 50 studies that reported it, and was absent in 39% of studies ($n=32$). The most frequent target muscles were the wrist and finger extensors, followed by the flexors; in lower-limb studies, the ankle dorsiflexors, as in post-stroke foot-drop correction systems [17]. Devices were varied commercial stimulators (e.g., Hasomed RehaStim, Compex Motion, g.tec), unnamed in 16 studies. Table II summarizes the FES parameters.

TABLE II
REPORTED FES STIMULATION PARAMETERS

Parameter	Reported, n (%)	Median	Most frequent
Stimulation frequency	48 (59%)	35 Hz	n/a
Pulse width	50 (61%)	250 μ s	n/a
Stimulation type	78 (95%)	n/a	Transcutaneous ($n=76$)
Device	66 (80%)	n/a	Varied

F. Intervention Protocols

The number of sessions was reported in 84% of studies ($n=69$), with a median of 12. Session duration had a median of 60 min, reported in 71% ($n=58$).

G. Outcomes and Results

The Fugl-Meyer Assessment for the Upper Extremity (FMA-UE) was the most used primary motor outcome ($n=21$

studies); the Action Research Arm Test (ARAT) appeared as a secondary measure in several additional studies.

Clinical outcome measurement varied with the developmental stage of the studies. Twenty-nine percent ($n=24$) assessed only BCI system performance metrics, such as classification accuracy and signal quality; many are engineering or feasibility studies, in which technical performance is the outcome pertinent to that stage. Among the 58 studies that assessed clinical outcomes, 46 (56% of the full corpus) reported statistically significant improvements in at least one primary motor measure; 12 (15%) reported mixed or inconclusive results; and no study reported exclusively null or negative motor findings (the boundary between positive and mixed depends on the criterion of significant improvement in at least one primary motor measure, and the absence of negative studies is consistent with a publication filter).

Formal efficacy evaluation was infrequent. Of the 82 studies, 26 (32%) had a separate control group, but only 17 statistically compared the groups on a clinical outcome; of the remaining 9, 7 reported within-group results only and 2 compared only a non-clinical outcome (decoding accuracy) against healthy controls. In those 17 studies, BCI-FES exceeded the control in 10 and did not differ significantly in the remaining 7. As two of the ten favorable reports describe the same trial (Krueger et al., 2022 and 2024), superiority corresponds to 9 of 16 independent cohorts.

This figure is a vote count of statistical significance across small and heterogeneous studies (median $n=7$) and should be read as a descriptive signal. Most reported significance only, without quantifying effect magnitude, which therefore could not be pooled; the absence of significance may reflect low statistical power, and publication bias favoring positive outcomes cannot be excluded.

IV. DISCUSSION

A. Technical Profile and Methodological Fragmentation

The included studies describe a technically convergent field: non-invasive EEG capturing sensorimotor signals (predominantly at C3, C4, and Cz) to drive transcutaneous FES of the upper limb in stroke survivors. The predominance of the upper limb (78%) follows the clinical profile of the dominant population, since stroke survivors (72%) most commonly present hand and arm deficits. SCI accounted for 28% of studies, some using intracortical recording to restore grasp and reach, as in the demonstration by Ajiboye et al. [14]. The corpus thus ranges from non-invasive feasibility systems to invasive functional-restoration interfaces, and from designs in which the neural signal continuously modulates stimulation to those in which a classifier's discrete output triggers a pre-configured FES pattern, a heterogeneity worth weighing when interpreting aggregate counts under the single label "BCI-FES".

On this technical convergence rests a methodological fragmentation. Motor imagery (MI, 48%) and movement execution or attempt (ME, 38%) appear in similar proportions and rarely

with explicit differentiation, although they engage distinct neural circuits [95] (MI activates cortico-cortical motor-simulation networks; ME recruits residual corticospinal pathways) and generate different EEG signatures in topography, latency, and frequency content. In plegic patients, the boundary between imagery and attempted movement is clinically thin, since the attempt may not produce observable movement. Treating neurophysiologically distinct paradigms as equivalent, together with the variability in FES parameter reporting discussed below, makes it difficult to pool results and compare protocols that, on the surface, appear similar.

B. Efficacy: A Narrow Evidence Base

Only 26 of the 82 studies (32%) included a separate control group and, of these, 17 conducted a between-group statistical comparison on a clinical outcome, with superiority reported in favor of BCI-FES in 10 (59% of those tested; 38% of controlled studies), or 9 of 16 independent cohorts (56%) after merging the two reports of the same Krueger trial. This result accompanies the positive randomized trials in the corpus: Biasiucci et al. [7] documented lasting arm motor recovery after stroke, Krueger et al. [15] attributed the gain to Hebbian plasticity induced by the temporal coincidence of intention and stimulation, and Chen et al. [16] reproduced the effect in chronic stroke; in SCI, the randomized pilot trial of Osuagwu et al. [19] pointed in the same direction. The meta-analyses of Ren et al. [9] and Liang et al. [10] estimate a moderate effect magnitude in upper-limb rehabilitation after stroke, compatible with the proportion observed here.

The broad reading, however, qualifies this signal. The meta-analyses synthesize only the controlled stroke subset, whereas the full mapping shows that such trials are a small fraction of the published work: most controlled studies had a small sample (median $n=7$), the 15 randomized trials (18%) follow the same size pattern, and 68% of studies included no comparator at all. The efficacy evidence remains preliminary: the 59% proportion results from vote counting of significance in small studies, which limits its interpretation as a measure of efficacy. Effect magnitude was rarely quantified: most controlled studies reported statistical significance only, and few reported a standardized effect size (for example, Biasiucci et al. [7], Cohen's $d = 1.03$; Krueger et al. [15], $\eta^2 = 0.57$; Ramirez-Nava et al. [86], Cohen's $d = 0.99$ on the Fugl-Meyer), which prevents a pooled magnitude estimate. Publication bias favoring positive outcomes also cannot be excluded.

C. FES Parameter Reporting: A Documented Variability

A contrast stands out in FES reporting. While nearly all studies described the mechanism by which the BCI triggers stimulation, the frequency was absent in 41% of articles and the pulse width in 39%, proportions that remain relevant even among the controlled designs. The studies thus document how they integrate BCI and FES, but not always with which stimulation parameters. The omission has a practical consequence: a recent meta-analysis associated, in exploratory subgroup analyses, specific ranges with larger motor effects (frequency

>50 Hz, pulse width 150 μ s), although the between-subgroup comparison did not reach statistical significance [10], so that positive results obtained without recording these settings become difficult to reproduce or compare.

D. Scope of the Evidence: Motor BCIs and Cognitive Outcomes

Within the studies included in this review, FES control relied on sensorimotor-cortex signals; frontal-pole electrodes (Fp1/Fp2) rarely served as the primary source (8 studies, 10%), and cognitive outcomes appeared only as secondary endpoints (13 studies, 16%), never as a primary target. Some prefrontal attention-based systems (Fp1) drive rehabilitation through a powered pedaling robot rather than FES [12], [13]; they fall outside this review but point to a prefrontal attention-based approach (Fp1/Fp2) not yet combined with FES in the mapped corpus.

E. Limitations

The acknowledged limitations include: single-reviewer screening and extraction, partly mitigated by mandatory verbatim quotation for six critical fields; the absence of a formal methodological quality assessment, consistent with scoping-review methodology [11]; a search restricted to three bibliographic databases, without grey literature or trial registries; probable publication bias favoring positive outcomes; and cohort overlap among some publications from the same group, which inflates participant totals relative to the number of independent samples.

F. Implications and Future Directions

Three directions emerge from this mapping. First, the variability in FES parameter reporting (Section IV-C) supports the adoption of a minimum reporting set for BCI-FES studies, a checklist containing stimulation frequency, pulse width, intensity, target muscles, and session regimen; without these items, protocols cannot be compared or replicated, and the ranges associated with larger effects by Liang et al. [10] become actionable only if these parameters are systematically recorded. Second, although technical feasibility is suggested mainly in chronic upper-limb stroke, with sparser demonstrations in other populations, the consolidation of therapeutic evidence depends on larger randomized trials, particularly in SCI, the lower limb, and the acute phase, underrepresented in the corpus and relevant for populations with gait deficits. Third, the prefrontal attention-based BCI approach (Fp1/Fp2), already explored with robotic effectors [12], [13], has not yet been combined with FES within the corpus mapped here; given the role of attention in engagement and in the effective training dose during rehabilitation, this integration represents a promising and as-yet-unexplored direction.

V. CONCLUSION

Eighty-two primary studies published between 2010 and 2025 characterize a field with a consistent technical profile: non-invasive EEG, motor-intention paradigms, and surface

FES targeting predominantly the upper limb in stroke survivors. Among the studies that formally compared BCI-FES to a control, superiority was reported in about half of the between-group comparisons (10 of 17; 9 of 16 independent cohorts), a proportion qualified by small sample sizes (median $n=7$); randomized trials total 15 studies (18%). Substantial variability in FES parameter reporting is documented across study designs, and 29% of studies assessed system performance without measuring any clinical motor outcome. The corpus mapped here suggests technical feasibility concentrated in chronic upper-limb stroke, with sparser demonstrations in other populations and configurations and mostly in small studies, while protocol standardization and the replication of larger controlled evidence remain infrequent.

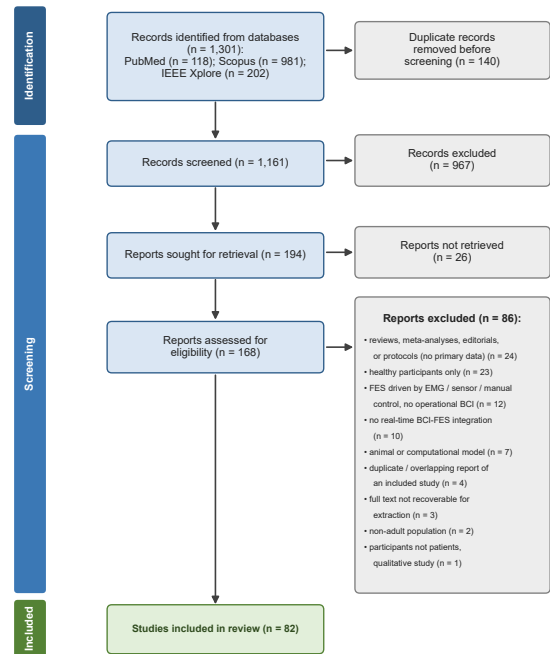


Fig. 1. PRISMA-ScR flow diagram of study selection (1,301 records identified; 1,161 after deduplication; 194 sought; 168 full texts assessed; 82 included).

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TABLE III

INCLUDED STUDIES (N=82): CHARACTERISTICS AND OUTCOMES. COND.: STROKE; SCI (SPINAL CORD INJURY). DESIGN: RCT; PRCT (PILOT RCT); NRC (NON-RANDOMIZED CONTROLLED); SC (SINGLE CASE, N=1); FP (FEASIBILITY/PILOT); CS (CASE SERIES); OTH (OTHER). PARAD.: MI (MOTOR IMAGERY); ME (EXECUTION/ATTEMPT); ERD; ATT (ATTENTION); P300. FES: FREQUENCY (Hz) / PULSE WIDTH (μ s). CTRL: SEPARATE CONTROL GROUP. OUTCOME: POS (POSITIVE); MIX (MIXED); NCO (NO CLINICAL OUTCOME); "SUP." INDICATES BETWEEN-GROUP SUPERIORITY FAVORING BCI-FES; "N.S." INDICATES A NON-SIGNIFICANT BETWEEN-GROUP DIFFERENCE.

Ref	Study	Cond.	Design	Parad.	FES Hz/ μ s	n	Ctrl	Outcome	
[20]	Tan et al.	Stroke	NRC	MI	25/250	9	No	NCO	
[21]	Tam et al.	Stroke	CS	MI	40/300	5	No	MIX	
[22]	Tan et al.	Stroke	FP	MI	25/250	9	No	NCO	
[17]	Do et al.	Stroke	SC	ME	NR/NR	1	No	NCO	
[23]	Takahashi et al.	Stroke	SC	ME	50/1200	1	No	POS	
[24]	Liang et al.	Stroke	FP	MI	NR/NR	8	Yes	NCO	
[25]	Ono et al.	Stroke	SC	ME	20/100	1	No	POS	
[26]	Pedrocchi et al.	Stroke	FP	P300	25/225	5	No	NCO	
[27]	Rohm et al.	Stroke	SCI	SC	MI	16/NR	1	No	POS
[28]	Tam et al.	Stroke	FP	MI	NR/NR	5	No	NCO	
[29]	Wang et al.	Stroke	NRC	MI	NR/NR	10	Yes	POS	
[30]	Chen et al.	Stroke	NRC	MI	NR/NR	10	Yes	NCO	
[31]	Ibáñez et al.	Stroke	NRC	ME	30/250	5	No	NCO	
[32]	King et al.	Stroke	SCI	SC	ME	NR/NR	1	No	NCO
[33]	Li et al.	Stroke	RCT	MI	60/150	15	Yes	POS, sup.	
[34]	Mukaino et al.	Stroke	SC	ERD	NR/NR	1	No	POS	
[35]	Young et al.	Stroke	SC	ME	NR/NR	1	No	POS	
[36]	Zhang et al.	Stroke	NRC	MI	NR/NR	8	Yes	NCO	
[37]	Chung et al.	Stroke	pRCT	Att	35/250	10	Yes	POS, n.s.	
[38]	Jiang et al.	Stroke	CS	MI	NR/NR	4	No	NCO	
[39]	King et al.	Stroke	SCI	SC	ME	NR/NR	1	No	NCO
[40]	McCrannon et al.	Stroke	NRC	ME	NR/NR	9	No	POS	
[41]	Vuckovic et al.	Stroke	SCI	CS	MI	30/300	2	No	MIX
[42]	Jure et al.	Stroke	FP	MI	NR/NR	NR	No	NCO	
[43]	Kim et al.	Stroke	RCT	Att	60/150	30	Yes	POS, sup.	
[18]	Mrachacz-Kersting et al.	Stroke	NRC	ME	NR/1000	22	Yes	POS	
[19]	Osugawu et al.	Stroke	SCI	pRCT	ME	26/200	12	Yes	POS, sup.
[14]	Ajiboye et al.	Stroke	SCI	SC	ME	12.5/100	1	No	POS
[44]	Bayon-Catalayud et al.	Stroke	SCI	OTH	ME	NR/NR	1	No	POS
[45]	Friedenberg et al.	Stroke	SCI	SC	MI	20/500	1	No	POS
[46]	Ibáñez et al.	Stroke	SC	ME	30/350	4	No	POS	
[47]	Irimia et al.	Stroke	CS	MI	NR/NR	2	No	POS	
[48]	Marquez-Chin et al.	Stroke	SC	ERD	NR/NR	1	No	POS	
[49]	Poboroniuc et al.	Stroke	CS	MI	50/255	3	No	MIX	
[7]	Biasiucci et al.	Stroke	RCT	ME	23/255	27	Yes	POS, sup.	
[50]	Cho et al.	Stroke	NRC	MI	50/300	7	No	POS	
[51]	Gant et al.	Stroke	SCI	NRC	MI	NR/200	14	Yes	NCO
[52]	Hashimoto et al.	Stroke	NRC	ME	33.3/300	4	No	MIX	
[53]	Irimia et al.	Stroke	CS	MI	NR/300	5	No	POS	
[54]	Lupu et al.	Stroke	NRC	MI	NR/NR	7	No	NCO	
[55]	Remsik et al.	Stroke	RCT	ME	60/150	21	Yes	POS, n.s.	
[56]	Trincado-Alonso et al.	Stroke	SCI	NRC	ME	40/350	4	No	MIX
[57]	Bockbrader et al.	Stroke	SCI	SC	ME	NR/NR	1	No	POS
[58]	Cho et al.	Stroke	CS	MI	50/300	2	No	POS	
[59]	Jovanovic et al.	Stroke	SC	ERD	40/250	1	No	POS	
[60]	Remsik et al.	Stroke	RCT	ME	60/150	21	Yes	POS	
[61]	Selfslagh et al.	Stroke	SCI	CS	MI	30/300	2	No	POS
[62]	Choi et al.	Stroke	NRC	MI	NR/NR	8	No	NCO	
[63]	Chung et al.	Stroke	pRCT	Att	35/250	25	Yes	POS, sup.	
[64]	Miao et al.	Stroke	RCT	ME	NR/300	16	Yes	POS, n.s.	
[65]	Murovec et al.	Stroke	FP	MI	50/350	3	No	POS	
[66]	Sebastián-Romagosa et al.	Stroke	FP	MI	50/300	51	No	POS	
[67]	de Castro-Cros et al.	Stroke	FP	MI	NR/NR	16	Yes	NCO	
[68]	Broniera Junior et al.	Stroke	SCI	SC	MI	35/123	1	No	NCO
[69]	Cajigas et al.	Stroke	SCI	SC	MI	NR/NR	1	No	POS
[16]	Chen et al.	Stroke	RCT	MI	25/NR	32	Yes	POS, sup.	
[70]	Jovanovic et al.	Stroke	SCI	FP	ERD	40/250	8	No	NCO
[71]	Jovanovic et al.	Stroke	SCI	FP	ERD	40/250	5	No	POS
[72]	Jovanovic et al.	Stroke	SCI	FP	ERD	40/250	8	No	POS
[73]	Remsik et al.	Stroke	NRC	ME	NR/NR	16	No	MIX	
[74]	Sinha et al.	Stroke	NRC	ME	60/150	20	Yes	POS	
[75]	Zulauf-Craja et al.	Stroke	SCI	FP	ME	33/250	8	No	NCO
[76]	Hashimoto et al.	Stroke	CS	ME	33.3/300	4	No	MIX	
[77]	Kim et al.	Stroke	NRC	OTH	NR/NR	34	No	NCO	
[78]	Krueger et al.	Stroke	RCT	ME	NR/NR	12	Yes	POS, sup.	
[79]	Kumari et al.	Stroke	OTH	ME	33/200	30	Yes	NCO	
[80]	Remsik et al.	Stroke	OTH	ME	60/150	3	No	NCO	
[81]	Wang et al.	Stroke	NRC	MI	30/300	7	Yes	POS, n.s.	
[82]	Zhan et al.	Stroke	RCT	MI	NR/NR	24	Yes	POS, n.s.	
[83]	Bigoni et al.	Stroke	SC	ME	NR/NR	1	No	POS	
[84]	Canillo-Negrete et al.	Stroke	SCI	FP	MI	30/300	6	No	MIX
[85]	Fioren et al.	Stroke	SCI	NRC	MI	20.5/400	2	No	NCO
[86]	Ramirez-Nava et al.	Stroke	NRC	P300	30/375	14	Yes	POS, sup.	
[87]	Sebastián-Romagosa et al.	Stroke	NRC	MI	50/350	25	No	POS	
[88]	Brunner et al.	Stroke	RCT	MI	50/300	40	Yes	MIX, n.s.	
[89]	Herring et al.	Stroke	SCI	SC	ME	NR/NR	1	No	POS
[15]	Krueger et al.	Stroke	RCT	ME	NR/NR	12	Yes	POS, sup.	
[90]	Kumari et al.	Stroke	SCI	FP	ME	33/230	10	Yes	MIX, n.s.
[91]	Luo et al.	Stroke	RCT	MI	NR/NR	64	Yes	POS, sup.	
[92]	Bastos-Filho et al.	Stroke	SC	MI	45/250	1	No	MIX	
[93]	Bastos-Filho et al.	Stroke	SC	MI	20/200	1	No	MIX	
[94]	Mansour et al.	Stroke	FP	ME	NR/NR	7	No	POS	

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